

Task 3. Validation

Objective: surprisingly before creating a ML model we will jump to the last step of the ML model building process. The objective of this task is to evaluate whether a given forecast is genuinely useful and statistically superior to simple alternatives. As for a given forecast you can use any baseline. It is best to come up with a few simple baselines. The real complication comes with the question “Is model A statistically the best model?”. In order to answer it you will need to complete several steps. You should focus on the validation process: how to design a fair out-of-sample evaluation, how to choose appropriate quality metrics, how to detect leakage or unstable results, and how to decide whether one forecast is statistically better than another. This is important because in high-frequency trading, small apparent improvements can easily be caused by data leakage, market regime dependence, transaction costs, overlapping observations, or multiple testing. A forecast that looks good on average may still be useless or misleading in live trading conditions. Highly recommend to use as a sense check a random filter as one of the baselines.

Data: <https://drive.google.com/file/d/1FpLBVKViVhvrFXw1vgBL743Y1IFWDwFD/view?pli=1>

From 2025-11-01 to 2026-04-28:

<https://drive.google.com/file/d/1XmxRsElei-vE8Gc5tkKs2wH4FJVRTevS/view>

Deliverables: Jupyter Notebook

- Clean, documented code with markdown explanations
- Executive summary with actionable insights and limitations of the analysis & data caveats

Assessment:

- Technical execution: data handling, code quality, visualization, etc.
- Analytical depth: casual relationship, pattern discovery, hypothesis generation, critical reflection, no data leakage etc.
- Communication & Deliverables: oral presentation, notebook structure, executive summary, reproducibility

Task 3.1 Official Scoring, Sanity Checks, and Basic Validation

- Implement the official markout and score calculation. Highly recommended to revisit how markout, PnL, score should be calculated and what additional limitations we have. You should think not only about limitations explicitly stated in the document but more about real-live limitations of any trading system.
- Think of quality metrics of your baseline filters
- Validate baselines in terms of markout, score and quality metrics you chose. Is there any correlation between filter quality and markout\score?

Task 3.2 Robustness, Statistical Superiority, and Overfitting Control

Here we will go through classical time-series validation techniques.

- Start with the simplest ones - walk-forward backtesting and rolling-window backtesting.
- Move to more advanced ones like purged and embargoed cross-validation and combinatorial purged cross-validation. The list is not exhaustive. Additional ideas are encouraged.
- Move to simple statistical tests. For example, think of such a test that answers the following question: “the candidate filter has no better expected score than the baseline filter.” Think about properties of the data you have and appropriate statistical tests.
- Move to bootstrapping. You have time series data. Can we use classical bootstrap here? How can we say that our filters work well with the help of bootstrapping?

Task 3.3 Advanced Validation Techniques (no hints here)

- How can you tell that model A predictions/forecasts are statistically better than model B?
 - What metric?
 - What statistical test given multiple time-series comparisons?
 - What if you have a set of N models? Can we choose one model's predictions that are superior in some terms than others?
- If the forecast were deployed live tomorrow, what monitoring system would you require to detect degradation, instability, or hidden leakage?
- Should this forecast be accepted, rejected, or kept under further observation? Justify the decision.